Yilin Zhang

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Education

Carnegie Mellon University Pittsburgh, PA
M.S. in Intelligent Information Systems, Language Technologies Institute *Dec 2025*
- Advisor: Prof. Sherry Tongshuang Wu
- GPA: 4.05/4.3

University of Waterloo Waterloo, Canada
B.S. in Computer Science (Honours, Co-op), Minor in Combinatorics and Optimization *Jun 2023*
- GPA: 92.72/100, Graduated with Dean's Honours (highest distinction)

Publications

First-Author

[1] **How Should Planning Agents ReSHAPE Their Search Space?**
Yilin Zhang, *Xinran Zhao, Tong Chen, Ziming Yuan, Tongshuang Wu*
Under review | *Preliminary version at NeurIPS 2026 BiAlign Workshop* (**Oral**) [Paper] [Code]

[2] **Penalizing Length: Uncovering Systematic Bias in Quality Estimation Metrics**
Yilin Zhang, *Wenda Xu, Zhongtao Liu, Tetsuji Nakagawa, Markus Freitag*
COLM 2026 [Paper]

[3] **cAST: Enhancing Code Retrieval-Augmented Generation with Structural Chunking via Abstract Syntax Tree**
Yilin Zhang, *Xinran Zhao, Zora Zhiruo Wang, Chenyang Yang, Jiayi Wei, Tongshuang Wu*
Findings of EMNLP 2025 [Paper] [Code]

Co-Author

[4] **CodeScout: An Effective Recipe for Reinforcement Learning of Code Search Agents**
Lintang Sutawika, Aditya Bharat Soni, Bharath Sriraam R R, Apurva Gandhi, Taha Yassine, Sanidhya Vijayvargiya, Yuchen Li, Xuhui Zhou, ***Yilin Zhang***, *Leander Melroy Maben, Graham Neubig*
Under review [Paper]

[5] **Efficient Many-Shot In-Context Learning with Dynamic Block-Sparse Attention**
Emily Xiao, Chin-Jou Li, ***Yilin Zhang***, *Graham Neubig, Amanda Bertsch*
ACL 2025 [Paper]

[6] **RankLLM: A Python Package for Reranking with LLMs**
Sahel Sharifymoghaddam, Ronak Pradeep, Andre Slavescu, Ryan Nguyen, Andrew Xu, Zijian Chen, ***Yilin Zhang***, *Yidi Chen, Jasper Xian, Jimmy Lin*
SIGIR 2025 [Paper]

[7] **TherAIssist: Assisting Art Therapy Homework and Client-Practitioner Collaboration through Human-AI Interaction**
Di Liu, Jingwen Bai, Zhuoyi Zhang, ***Yilin Zhang***, *Zhenhao Zhang, Jian Zhao, Pengcheng An*
IMWUT (UbiComp 2025) [Paper]

[8] **Planar or Spatial: Exploring Design Aspects and Challenges for Presentations in Virtual Reality with No-coding Interface**
Liwei Wu, ***Yilin Zhang***, *Justin Leung, Jingyi Gao, April Li, Jian Zhao*
ISS 2024 [Paper]

Research Experience

Google People + AI Research (PAIR) Mountain View, CA
Research Lead | Advisor: Prof. Daphne Ippolito *Mar 2026 – Present*

- Leading a study of language models' individual-level, subjective theory of mind in social games.

Google Translate Research Mountain View, CA
Intern | Hosts: Zhongtao Liu, Wenda Xu | Advisor: Markus Freitag *May 2025 – Aug 2025*

- Led a study showing that MT quality-estimation metrics systematically penalize longer translations, and proposed mitigations [2].

CMU LTI Pittsburgh, PA
Research Assistant | Advisor: Prof. Sherry Tongshuang Wu *Aug 2024 – Present*

- Led a study showing that planning agents should revise users' specifications before planning, since initial requests often exclude plans users would accept [1].
- Led a study showing that AST-based structural chunking improves code retrieval-augmented generation, and released the **astchunk** library [3].

Waterloo HCI Waterloo, Canada
Research Assistant | Advisor: Prof. Jian Zhao *Jan 2023 – Apr 2024*

- Built a no-code authoring system for presentations in virtual reality [8].
- Built human–AI tools supporting art-therapy homework and client–practitioner collaboration [7].

Awards

Microsoft Research AI and the New Future of Work Award 2026
Co-wrote with Prof. Sherry Wu (PI): *A Collaboration Arena for AI Agents That Discover Hidden Team Needs*

Industry Experience

Google | *Software Engineer* Jan 2026 – Present
Amazon | *Software Engineering Intern* May 2022 – Dec 2022
Oracle | *Software Engineering Intern* Sep 2021 – Dec 2021

Academic Service

Program Committee: NeurIPS 2026 BiAlign Workshop 2026
Reviewer: ACL ARR (May 2025, Oct 2026), COLM 2026 2025 – 2026

Teaching

Teaching Assistant, CMU 11-442/642 Search Engines 2025
Head Teaching Assistant, Waterloo CS 240 Data Structures (5 semesters) 2020 – 2022
Teaching Assistant, Waterloo MATH 137/138 Calculus I/II 2020

Languages

Mandarin (native), English (fluent), Japanese (conversational), French (conversational)